

Class Imbalance in Histopathology: Injecting the Observed Prior Shift Overstates the TCGA-UT–BRACS Gap

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Abstract

At an imbalance ratio of $\rho = 100$, a skewed class prior alone costs 4.47 points of balanced accuracy on BRACS against 1.20 points on TCGA-UT, a gap of 3.27 points between two datasets fit under the same design, and the cause of that gap-within-a-gap is untested. This report tries to explain it with no new fits: for every stored balanced fit, it injects the paired long-tailed fit’s own realized class prior directly into the balanced model’s stored test probabilities, $\arg\max(\log p + \log \hat{\pi})$, and measures the resulting drop in patient-macro balanced accuracy, D_{sim} . Unlike a linear confusion-based score, this simulated damage automatically reflects each patch’s decision margin, so it should approximate the damage a genuine prior-reweighted refit produces, provided the frozen decision surface it starts from is not moved too far. A pre-registered gate checks exactly that: whether $D_{\text{sim}}(\rho = 100)$ falls inside the 95 percent interval of the already-fitted prior-only damage. The gate fails on both datasets, and by a wide and dataset-dependent margin: $D_{\text{sim}} = 10.55$ points on TCGA-UT against a fitted interval of $[0.87, 1.53]$, and $D_{\text{sim}} = 14.43$ points on BRACS against $[2.84, 6.30]$. No mechanism claim is drawn from this proxy, even though the cross-dataset difference in D_{sim} , 3.89 points with a 95 percent interval of $[1.12, 7.06]$, lands close to the 3.27-point target the two datasets’ fitted models actually show. A descriptive check restricts TCGA-UT’s balanced model to 500 random seven-class subsets and one subset chosen for maximal pairwise confusion, and finds a strong negative relationship between a subset’s margin and its simulated damage ($\rho_s = -0.55$, $p < 10^{-39}$); BRACS’s own margin and balanced accuracy nonetheless fall outside the range covered by every one of those subsets, so a narrower class set alone does not obviously place it back on the trend. The gate’s failure is traced to the injected decision rule itself: it moves a frozen decision surface rather than refitting one, so it overweights collateral confusion whenever within-dataset margins are moderate, and it does so more severely on TCGA-UT’s thirty-class design than the dataset’s own larger median margin would suggest.

1 Motivation

Two studies in this series measured how much of a skewed training class distribution’s damage to balanced accuracy comes from the shifted class prior alone, holding the classifier’s frozen ViT features fixed and separating the prior from the tail’s thin patch support (Queisler, 2026b,a). At the most severe ratio tested, $\rho = 100$, the prior-only arm P100 costs 1.20 points of test patient-macro balanced accuracy on TCGA-UT, with a 95 percent interval of $[0.87, 1.53]$, and 4.47 points on BRACS, with an interval of $[2.84, 6.30]$ (Queisler, 2026b,a). The two studies share every other design choice: the same allocator, the same three patient-disjoint splits, the same ten draws per split, and the same class-rank permutation scheme (Queisler, 2026e,d). The 3.27-point difference between them is therefore a difference in what the prior shift costs on each dataset, not a difference in how the prior shift is constructed, and nothing in the series so far has tested why it is that large.

A companion study on BRACS proposed an explanation and left it untested: that the damage of moving the class prior scales with how much the classes already overlap under balanced training, so a dataset whose classes are less separable pays more for the same nominal shift (Queisler, 2026d). The BRACS tail-assignment study lends this some support after the fact, since BRACS classes collapsed to almost no recall in the tail regardless of which class was rare, while TCGA-UT classes did not (Queisler, 2026f,g). Two candidate causes are confounded, however: BRACS also has far fewer classes than TCGA-UT, seven against thirty, and a smaller class count changes both the head-to-tail count ratio’s effect on decision boundaries and the base rate of confusable pairs. This report isolates the class-prior channel with no new fits and asks two questions: does the size of the 3.27-point gap follow from how the shifted prior interacts with each dataset’s own decision margins, and does that account survive a check restricted to seven classes on TCGA-UT, matching BRACS’s own class count.

2 Why not a linear confusion score

A natural alternative is a scalar score over classes, along the lines of $S = \frac{1}{K} \sum_c q_{c \rightarrow d} |\log \pi_c - \log \pi_d|$ for some confusion measure q , evaluated once per dataset and compared across the two. This report does not use that form, for six reasons. First, $|\cdot|$ discards direction, while a companion study on TCGA-UT found that the direction of a rank swap costs 1.11 points on its own, holding the pairwise separation fixed (Queisler, 2026h). Second, a companion tail-assignment study on BRACS found that headroom, a class’s own recall under balanced training, dominates its tail damage there (Queisler, 2026f), and a score built only from confusion mass q drops that term entirely. Third, q comes from balanced-arm argmax predictions and so discards the margin between the true class’s log-probability and its nearest competitor, and a prior shift acts on exactly that margin: two pairs with the same confusion rate can respond very differently to the same log-prior shift depending on how close their logits already are. Fourth, a linear score scales with $\log \rho$, so it cannot reproduce both a roughly 7-fold BRACS-to-TCGA-UT damage ratio at $\rho = 10$ and a roughly 2.6-fold ratio at $\rho = 100$, since the two datasets’ damage curves begin rising, or set in, at different ratios (Queisler, 2026e,d). Fifth, with only two datasets, a pooled regression of one dataset-level score on one dataset-level damage cannot separate the score’s explanatory power from any other dataset-level difference between BRACS and TCGA-UT. Sixth, and most directly, the pairwise-confusion machinery this kind of score is built from was already checked against fitted damages on TCGA-UT and did not rank three known class orders correctly (Queisler, 2026i), so reusing the same ingredients across datasets carries no presumption of working better here.

The alternative pursued instead needs no calibration and no new fits: inject the prior directly into the balanced model’s own stored decisions and measure what happens, so that margins, direction, headroom, class count, and the nonlinearity of when damage sets in all come through the classifier’s actual output rather than through a hand-built summary.

3 Design

For a stored balanced fit $r1$ of one (split, draw) and its paired long-tailed fit $r\rho$ of the same split and draw, this report reuses $r1$ ’s stored test probabilities p unchanged and injects $r\rho$ ’s realized class counts n_c as a log-prior shift,

$$\hat{\pi}_c = \frac{n_c}{\sum_{c'} n_{c'}}, \quad \hat{y} = \operatorname{argmax}_c (\log p_c + \log \hat{\pi}_c), \quad (1)$$

and reports $D_{\text{sim}}(\rho)$, the resulting drop in pooled test patient-macro balanced accuracy against $r1$ ’s own predictions. Equation (1) is the mirror image of the post-hoc logit adjustment a companion study already fits and validates, which subtracts a fit’s own implicit prior from

its logits to partially undo a prior shift (Queisler, 2026b; Menon et al., 2021); here the same adjustment adds a prior shift to the balanced fit instead of removing one from the shifted fit, and the source of p is always r1 rather than the arm being adjusted. No solver is refit: r1 and $r\rho$ are reused exactly as stored, addressed against the two datasets’ own prevalence and prior-versus-support outputs (Queisler, 2026e,d,b,a). Arms $Q\rho$ are computed for $\rho \in \{2, 5, 10, 20, 50, 100\}$ on both datasets, at three patient-disjoint splits and ten draws per split, matching the paired r1/ $r\rho$ design of the prior-versus-support studies. A validation-temperature-scaled variant $QT\rho$, injecting the same prior into r1’s calibrated rather than raw probabilities, is computed identically as a sensitivity check.

3.1 Pre-registered gate and hypotheses

Before any injected arm is inspected, this report fixes a validity gate, a primary hypothesis, and two descriptive checks.

- **Gate G** (validity, per dataset): $D_{\text{sim}}(\rho = 100)$ ’s point estimate falls inside the already-fitted P100 damage’s 95 percent interval, $[0.87, 1.53]$ on TCGA-UT and $[2.84, 6.30]$ on BRACS (Queisler, 2026b,a). On failure, no further hypothesis is treated as a mechanism claim.
- **H1** (primary): the cross-dataset gap $\Delta_{\text{sim}} = D_{\text{sim},\text{BRACS}}(100) - D_{\text{sim},\text{TCGA-UT}}(100)$, with a 95 percent interval built by independently resampling each dataset’s own bootstrap distribution, so that no shared random draw pairs the two datasets’ replicates. H1 is read as support for the prior-channel account if Δ_{sim} agrees with the fitted 3.27-point gap within 1 point.
- **H2** (onset, descriptive): the $D_{\text{sim}}(\rho)$ curve is plotted against the fitted $P(\rho)$ curve at every ρ , and the onset ratio, the first ρ whose interval clears zero, is reported for each.
- **H3** (class-count control, descriptive): TCGA-UT’s r1 probabilities are restricted to seven-class subsets, renormalized over the subset, and injected with a synthetic $\rho = 100$ profile under a random class-to-rank assignment, using only the subset’s own test patches; 500 random subsets and one subset built by greedily growing the most pairwise-confused group of seven classes are scored this way and plotted against each subset’s own balanced accuracy and median margin, with BRACS’s own point placed on the same axes. A restricted softmax over a subset of an already-fit thirty-class model is not the same object as a model trained on those seven classes from scratch, so H3 is read qualitatively, not as a formal test.

Three outcome labels are fixed along with the gate: *margin_explains_prior_gap* if the gate passes and H1 supports the fitted gap, *partial* if the gate passes but H1 misses by more than one point, and *proxy_invalid* if the gate fails on either dataset.

4 Results

All 360 injected fits ($2 \text{ datasets} \times 6 \text{ ratios} \times 3 \text{ splits} \times 10 \text{ draws}$) and their validation-temperature-scaled sensitivity variant converged with no missing records; Table 1 reports the full curve for both.

4.1 Gate G fails on both datasets

$D_{\text{sim}}(\rho = 100)$ falls far outside the already-fitted P100 interval on both datasets, and the two failures are not the same size. On TCGA-UT, $D_{\text{sim}}(100) = 10.55$ points, roughly seven times the

fitted point estimate of 1.20 and outside the fitted interval $[0.87, 1.53]$ by a factor of about seven at the nearer bound. On BRACS, $D_{\text{sim}}(100) = 14.43$ points against a fitted point estimate of 4.47 and an interval of $[2.84, 6.30]$, an overshoot of a little over three times at the nearer bound. Gate G fails on both datasets, so H1 is not read as a mechanism claim, and every number in this section is reported to characterize the proxy’s failure rather than to support the prior-channel account it was built to test.

4.2 H1: the cross-dataset gap, reported without a mechanism claim

Because the gate fails, H1’s outcome is descriptive. The simulated gap is $\Delta_{\text{sim}} = D_{\text{sim, BRACS}}(100) - D_{\text{sim, TCGA-UT}}(100) = 14.43 - 10.55 = 3.89$ points, with an independently-resampled 95 percent interval of $[1.12, 7.06]$, which covers the fitted models’ 3.27-point gap. This agreement does not rehabilitate the proxy: the two datasets overshoot their own fitted damage by different multiplicative factors, about $7.0\times$ on TCGA-UT against about $3.2\times$ on BRACS, so a difference that lands near the right value does so despite the two absolute values being wrong by different amounts, not because the errors cancel by a shared correction factor. Reading Δ_{sim} as support for the prior-channel account would require the gate to have passed first; it did not, and this number is reported only so that the coincidence is on the record rather than quietly omitted.

4.3 H2: the simulated curve separates from the fitted curve early

ρ	TCGA-UT				BRACS			
	D_{sim} (pp)		Fitted P (pp)		D_{sim} (pp)		Fitted P (pp)	
2	0.29	(0.17 to 0.39)	0.10	(−0.00 to 0.21)	1.53	(0.48 to 2.65)	0.98	(0.00 to 2.18)
5	1.52	(1.26 to 1.78)	0.30	(0.09 to 0.51)	4.84	(3.05 to 6.66)	2.13	(0.89 to 3.48)
10	3.06	(2.67 to 3.44)	0.49	(0.25 to 0.73)	7.55	(5.31 to 9.90)	2.77	(1.39 to 4.28)
20	4.97	(4.42 to 5.49)	0.73	(0.45 to 1.02)	10.10	(7.58 to 12.73)	3.68	(2.16 to 5.34)
50	7.98	(7.27 to 8.69)	0.97	(0.65 to 1.28)	12.73	(10.01 to 15.64)	4.10	(2.57 to 5.73)
100	10.55	(9.72 to 11.38)	1.20	(0.87 to 1.53)	14.43	(11.71 to 17.42)	4.47	(2.84 to 6.30)

Table 1: Simulated prior-injection damage $D_{\text{sim}}(\rho)$ against the already-fitted prior-only damage $P(\rho)$, both as the drop in pooled test patient-macro balanced accuracy from the balanced arm, with 95 percent bootstrap intervals over three splits and ten draws per split. The two curves agree at $\rho = 2$ on both datasets and separate at every larger ratio.

Figure 1 plots both curves for each dataset. At $\rho = 2$ the two curves agree within one point on both datasets, with D_{sim} and the fitted P sharing an onset ratio of 2 on BRACS. On TCGA-UT the fitted curve rises so slowly that its own onset, the first ratio whose interval clears zero, is not reached until $\rho = 50$, whereas D_{sim} ’s interval clears zero already at $\rho = 2$. From $\rho = 5$ onward the two curves separate on both datasets and never rejoin: D_{sim} accelerates roughly log-linearly with ρ on both datasets, while the fitted curve stays comparatively flat, so the size of the simulated curve’s overshoot itself grows with ρ rather than sitting at a fixed offset.

4.4 H3: a class-count control does not bring BRACS back onto the trend

Restricting TCGA-UT’s balanced model to 500 random seven-class subsets of its 30 classes, and injecting a synthetic $\rho = 100$ profile with a random class-to-rank assignment within each subset, produces a strong negative relationship between how hard a subset already is and how much the injection damages it: Spearman correlations of -0.53 ($p < 10^{-37}$) between a subset’s own balanced accuracy and its simulated damage, and -0.55 ($p < 10^{-39}$) between its median margin and its simulated damage. Both are shown in Figure 2. A subset built by greedily growing the

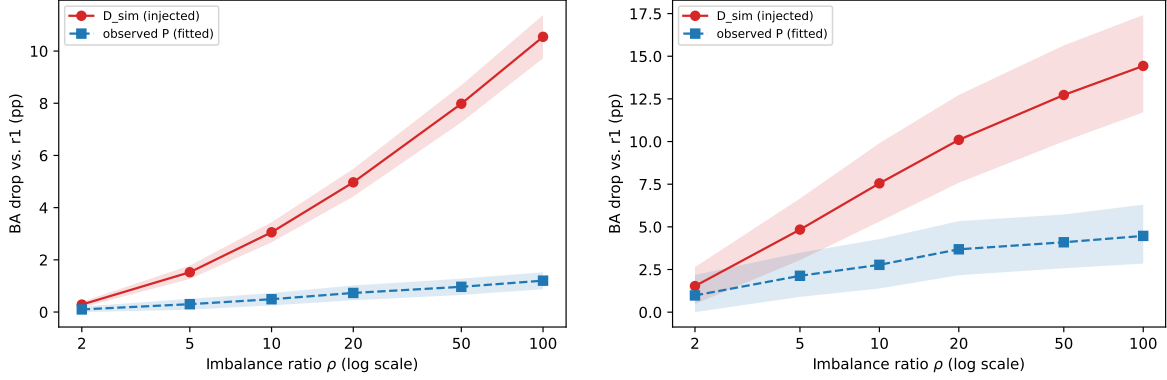


Figure 1: The simulated damage $D_{sim}(\rho)$ (red, injected into the frozen balanced decision surface) against the already-fitted prior-only damage $P(\rho)$ (blue, from a genuine reweighted refit), on TCGA-UT (left) and BRACS (right). Points are pooled over three splits and ten draws per split; bands are 95 percent bootstrap intervals. The two curves track each other only at the smallest ratio.

group of seven classes with the largest pairwise confusion rate lands at a balanced accuracy of 58.5 percent, a median margin of 0.60 nats, and a simulated damage of 9.15 points, below the random subsets’ own minimum balanced accuracy of 67.4 percent and consistent with the same trend.

BRACS’s own balanced accuracy under its balanced arm, 39.6 percent, and its own median margin, 0.27 nats, both fall outside the entire range covered by the 500 random subsets, whose balanced accuracy spans 67.4 to 97.4 percent and whose median margin spans 0.96 to 5.93 nats, and outside the range reached by the single most-confused engineered subset as well. Restricting TCGA-UT to seven classes, even a deliberately hard seven, does not reproduce a task as separability-poor as BRACS’s native seven. A narrower class count on its own is therefore not sufficient to place BRACS back on the trend line these subsets trace; whatever makes BRACS’s classes harder to tell apart under balanced training is not simply a consequence of having fewer of them.

Figure 3 shows why a shared $\rho = 100$ profile produces such different damage on the two full datasets: BRACS’s test-patch margin distribution sits far closer to zero than TCGA-UT’s, so a much larger share of its patches fall within the log-count gap the injected prior introduces at any of the three marked ratios.

The validation-temperature-scaled sensitivity variant lowers the simulated damage but does not close the gate. Injecting the same prior into calibrated rather than raw probabilities gives $D_{sim,TS}(100) = 7.74$ points on TCGA-UT and 9.01 points on BRACS, 2.81 and 5.42 points below the raw-probability values respectively; both remain well outside their dataset’s fitted P_{100} interval. Calibration narrows the damage, consistent with temperature scaling spreading probability mass away from the single most likely class and so softening how much a fixed log-prior shift can move an argmax, but it does not narrow it enough to make the injection a valid stand-in for a genuine refit.

5 Discussion

Injecting a fitted dataset’s own realized class prior into its balanced model’s stored decisions was meant to give a zero-fit stand-in for the fitted prior-only damage, one that keeps the margins, the direction, and the class count a linear confusion score discards. It does keep all three, and its failure is informative about exactly which of them dominates: Section 4.3 shows the simulated

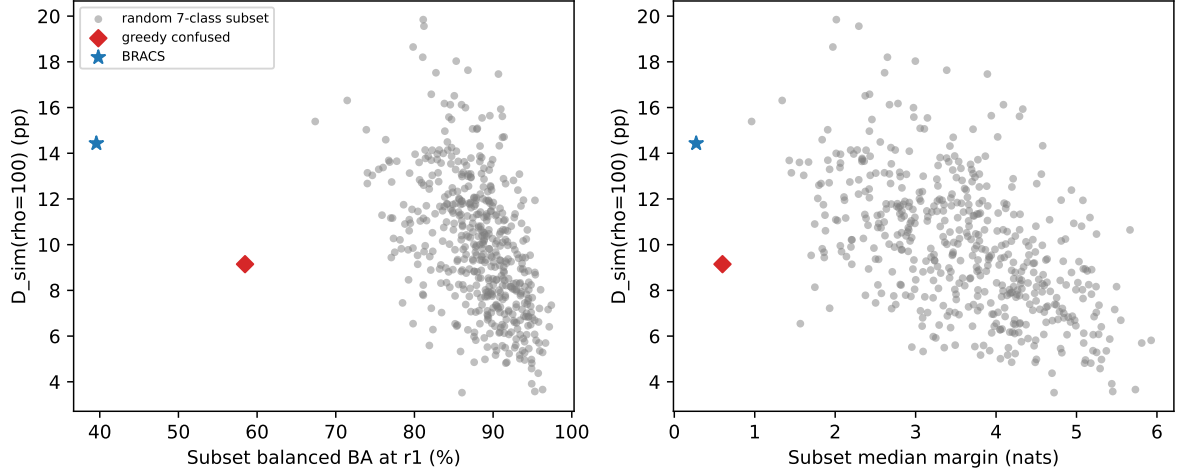


Figure 2: Simulated $\rho = 100$ damage against a seven-class subset’s own balanced accuracy (left) and median margin (right), for 500 random seven-class subsets of TCGA-UT’s 30 classes (grey), the single subset built from the seven most pairwise-confused classes (red diamond), and BRACS’s own balanced arm (blue star). BRACS falls outside the range every TCGA-UT subset covers on both axes.

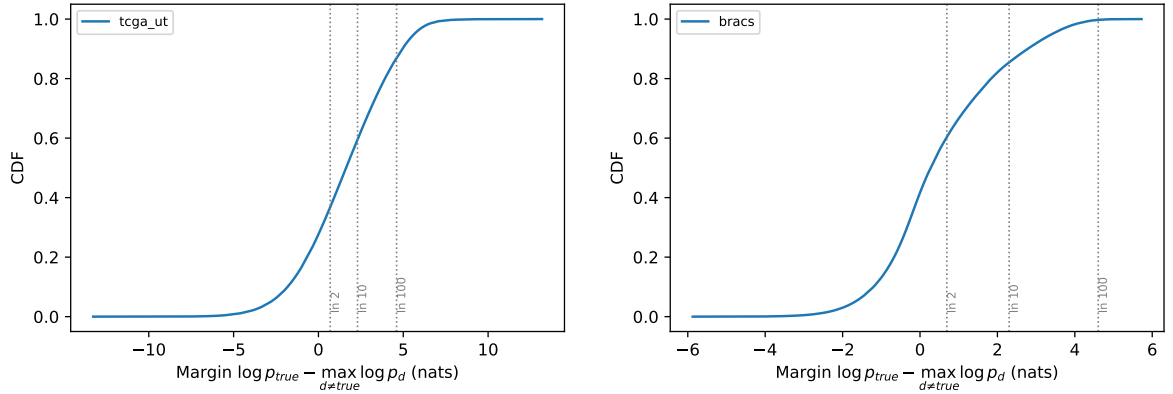


Figure 3: Cumulative distribution of the balanced arm’s test-patch margin, $\log p_{\text{true}} - \max_{d \neq \text{true}} \log p_d$, on TCGA-UT (left, median 1.63 nats) and BRACS (right, median 0.27 nats). Vertical lines mark $\ln \rho$ for $\rho \in \{2, 10, 100\}$: the share of patches with a smaller margin than a given line is the share directly exposed to a prior shift of that size.

and fitted curves agreeing only where the injected shift is small enough that few decisions move, and Section 4.4 shows the simulated damage tracking a subset’s own margin closely within one dataset. What the simulation does not do is reproduce the size of a genuine refit’s damage once the shift is large. A reweighted refit redistributes an entire decision surface to accommodate a new prior; the injection used here moves every decision independently against a surface that was never asked to accommodate one, so wherever a patch’s margin is smaller than the injected log-prior gap, the prediction flips regardless of whether a refit would have found a way to keep it. This is the same margin-sensitivity that motivated using simulation over a linear score in the first place, and it turns out to be strong enough that the simulation overshoots a genuine refit’s damage by a wide margin once ρ moves past the smallest ratio tested.

The size of that overshoot is itself dataset-dependent in a way median margin alone does not explain: TCGA-UT’s balanced arm has the larger median margin of the two datasets, 1.63 against BRACS’s 0.27 nats, yet TCGA-UT’s simulation overshoots its own fitted damage by roughly seven times against BRACS’s roughly three times. A median summarizes the typical patch, while the injected damage is set by whichever patches sit closest to the decision boundary in either direction; with thirty classes, TCGA-UT’s balanced model has many more pairwise boundaries than BRACS’s seven-class model does, and it is plausible that this gives more opportunities for some pair’s margin to be smaller than the injected gap even when the typical margin is comparatively large. This report does not test that account and does not extend it into a mechanism claim; it is offered only to keep the size of the gate’s failure from reading as a single fixed multiplier that a future report could simply divide out.

What the class-count control in Section 4.4 does show, without needing the gate to pass, is that BRACS’s balanced arm is harder, in both accuracy and margin terms, than any seven-class subset carved from TCGA-UT, including the one deliberately built to maximize pairwise confusion. If the 3.27-point fitted gap between the two datasets’ prior-only damage has a cause that generalizes past class count, it is a property of how BRACS’s seven classes overlap under balanced training that a same-sized slice of TCGA-UT’s thirty classes does not share, not an artifact of TCGA-UT’s larger label set diluting its own damage. That leaves the fitted gap’s cause open. The prior channel investigated here explains at most part of it, and the support channel, thin tail patch counts rather than a shifted prior, carries a comparable or larger share of BRACS’s total $\rho = 100$ damage in the companion prior-versus-support study (Queisler, 2026a) and remains untested for why it costs more on BRACS than on TCGA-UT. A companion patient-shortage study finds that most of the benefit of additional patients on a related task comes from a class-centre error rather than from added support directly (Queisler, 2026c); whether an analogous centre-error account explains the support channel’s dataset gap is the natural next candidate and is left open here.

6 Limitations

Gate G is a necessary check on one proxy’s absolute validity, not a diagnosis of what is wrong with it; failing it rules the proxy out for a mechanism claim without identifying which of the reasons discussed above is the operative one. H1’s numerical agreement with the fitted gap is reported for the record and is explicitly not evidence for the prior-channel account, since it survives only as a difference of two independently wrong quantities. H3’s restricted-softmax subsets are not retrained seven-class models: renormalizing an already-fit thirty-class softmax over seven columns changes what probability mass is available to redistribute in a way an actual seven-class refit would not, so the subset scores in Figure 2 should be read as a screen for whether class count alone can explain BRACS’s position, not as a measurement of what a seven-class TCGA-UT model would show. The greedy confused-class subset uses a single, un-bootstrapped pooled confusion matrix from the balanced arm and is one specific construction among many that could grow a hard subset; a different growth rule could plausibly find a harder one. All

results are conditional on the same design as the prior-versus-support studies: frozen Virchow2 features, multinomial logistic regression, the same three splits and ten draws per split, $G = 20$ patients per class on TCGA-UT and $G = 10$ on BRACS, and the class-rank permutation scheme of the prevalence curves (Queisler, 2026e,d,b,a).

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